ECE 310 Fall 2026
Lecture 4: Convolution and impulse response

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Learning Objectives

- Define the impulse response of a discrete-time system.
- Show how the response of LTI systems to input signals can be fully described by their impulse response.
- Define the convolution between two discrete-time signals and key properties of convolution.
- Compute the convolution between two discrete-time signals by hand.



Impulse Response

Impulse response

The *impulse response* $h[n]$ of a discrete-time system T is the response or output of the system to a unit impulse or Kronecker delta $\delta[n]$ input:

$$h[n] = T(\delta[n])$$

Let's consider a couple examples of impulse responses.

- $y_1[n] = x[n] - x[n-1]$
- $y_2[n] = \text{median}\{x[n], x[n-1], x[n-2]\}$



Signals as Combination of Impulses

- Consider $x[n] = \cos\left(\frac{\pi}{4}n\right)(u[n] - u[n-8]) = \left\{\underset{\uparrow}{1}, \frac{\sqrt{2}}{2}, 0, -\frac{\sqrt{2}}{2}, -1, -\frac{\sqrt{2}}{2}, 0, \frac{\sqrt{2}}{2}\right\}$.
  How can we write this signal as a linear combination of unit impulses?

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  How can we write this signal as a linear combination of unit impulses?
- Each sample in $x[n]$ is associated with a location k and value $x[k]$. Thus,

$$x[n] = \delta[n] + \frac{\sqrt{2}}{2}\delta[n-1] - \frac{\sqrt{2}}{2}\delta[n-3] - \delta[n-4] - \frac{\sqrt{2}}{2}\delta[n-5] + \frac{\sqrt{2}}{2}\delta[n-7]$$



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- In general, we may write any signal $x[n]$ as

$$x[n] = \sum_{k=-\infty}^{\infty} x[k]\delta[n-k]$$



System Response of Discrete-time Systems

Suppose we have a discrete-time system T with impulse response $h[n]$. How can we find the response of the system to arbitrary input signals?

$$\delta[n] \longrightarrow \boxed{T}$$

$$a\delta[n] \longrightarrow \boxed{T}$$

$$\delta[n-k] \longrightarrow \boxed{T}$$

$$\delta[n] + \delta[n-k] \longrightarrow \boxed{T}$$

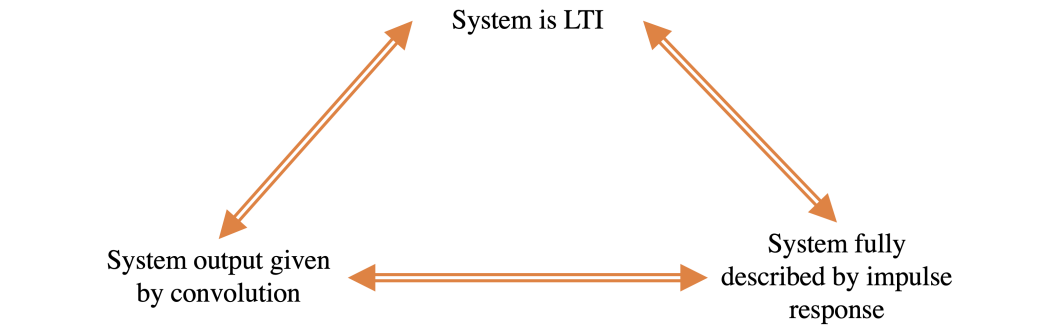
$$\sum_{k=-\infty}^{\infty} x[k]\delta[n-k] \longrightarrow \boxed{T}$$



Convolution

The system response $y[n]$ of an LTI system with impulse response $h[n]$ to input signal $x[n]$ is given by the *convolution* of $x[n]$ and $h[n]$, $y[n] = x[n] * h[n]$

$$y[n] = x[n] * h[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k] = \sum_{k-\infty}^{\infty} h[k]x[n-k] \tag{1}$$



Key Properties of Convolution and Impulse Response

- **Commutativity**:

$$x[n] * h_1[n] * h_2[n] = x[n] * h_2[n] * h_1[n] = h_1 * x[n] * h_2[n] = \cdots$$

- **Distributivity**:

$$x[n] * (h_1[n] + h_2[n]) = x[n] * h_1[n] + x[n] * h_2[n]$$

- **Length**: The convolution of length-N $x[n]$ with length-M $h[n]$ results in the length-$(N + M - 1)$ signal $y[n] = x[n] * h[n]$.

- **Start Point and End Points**: Suppose $x[n]$ has start and end indices (first and last non-zero value locations) n_0 and n_1, respectively; $h[n]$ has start and end indices k_0 and k_1; then, $y[n] = x[n] * h[n]$ will have start and end indices:

$$\begin{aligned} \text{Start Index} &= n_0 + k_0 \\ \text{End Index} &= n_1 + k_1 \end{aligned}$$



Key Properties of Convolution and Impulse Response

- **Identity**:
$$x[n] * \delta[n] = x[n]$$
- **Causality**: An LTI system with impulse response $h[n]$ is causal if and only if
$$h[n] = 0, \ n < 0.$$
- **Stability**: An LTI system with impulse response $h[n]$ is BIBO stable if and only if
$$\sum_{n=-\infty}^{\infty} |h[n]| < \infty.$$



Computing Convolution

The general steps to *convolve* two sequences:

1. Flip one of the signals, e.g. $h[-k]$.
2. Shift the flipped signal by n samples to the right, e.g. $h[-(k-n)] = h[n-k]$,
3. Multiply the unchanged signal with the flipped-and-shifted signal element-wise, e.g. $z_n[k] = x[k]h[n-k]$.
4. Sum up the product of the two signal to obtain the output at index n, e.g. $y[n] = \sum_{k=-\infty}^{\infty} z_n[k]$.



Computing Convolution: Shift-and-Overlap

Suppose we would like to convolve the following two sequences:

$$x[n] = \{\underset{\uparrow}{1}, 2, 0, -4, -1\}, \;\; h[n] = \{-2, \underset{\uparrow}{0}, 1\}$$

$$y[n] = x[n] * h[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k]$$

k	-2	-1	0	1	2	3	4	
$x[k]$			1	2	0	-4	-1	$y[n]$
$h[-(k+1)] = h[-1-k]$								
$h[-k]$		1	0	-2				
$h[-(k-1)] = h[1-k]$								
$h[-(k-2)] = h[2-k]$								
$h[3-k]$								
$h[4-k]$								
$h[5-k]$								

Computing Convolution: Matrix Method

Suppose we would like to convolve the following two sequences:

$$x[n] = \{\underset{\uparrow}{1}, 2, 0, -4, -1\}, \; h[n] = \{-2, \underset{\uparrow}{0}, 1\}$$



Computing Convolution: Applying Convolution Sum

Suppose we would like to convolve the following two sequences:

$$x[n] = \left(\frac{1}{2}\right)^n u[n],\ h[n] = \left(-\frac{3}{4}\right)^n u[n]$$



Cases for Computing Convolution

Three distinct cases for computing the convolution of two discrete-time signals:

1. Two finite-length signals
2. Two infinite-length signals
3. One finite-length signal and one-infinite-length signal



Computing Convolution: Finite and Infinite-length Sequences

Suppose we would like to convolve the following two sequences:

$$x[n] = \{-2, 0, \underset{\uparrow}{0}, 3, 0, 1\}, \; h[n] = e^{-n}u[n]$$



Extra Convolution Practice

Convolve the following two sequences:

$$x[n] = \{-3, 6, 2, \underset{\uparrow}{0}, 1, 1\}, \; h[n] = \{1, \underset{\uparrow}{-1}\}$$



Extra Convolution Practice

Convolve the following two sequences:

$$x[n] = u[n],\ h[n] = \left(\frac{2}{3}\right)^{n-2} u[n-4]$$

